KMT-2026-BLG-0521 / OGLE-2026-BLG-0058

Astrometric analysis — summary

Prepared 2026-09-06. Solution: `~/KMTdata/Results/v3_Final/IFfinal_260058_CTIO_.mat`

(variable `IFsys`). Code at commit `47c52cf` on branch `dev1`.

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1. Data

	BLG41	BLG01
exposures	19 133	19 981
after cuts	17 354	17 687
sources fitted	594	621
nights	1 985	2 009

KMT CTIO, 2016–2026, I band. Two **overlapping cut-outs** of the same star,

300 × 300 pix at 0.4″/pix (2′ × 2′). They share **no exposures**: the two fields

are observed alternately, a median of 12.2 min apart, on 1 947 common nights, so

they are independent measurements.

V-band frames exist for 2016–2022 (~1 340 per field) and are used only to check

the colour; they are not used in the astrometry.

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2. What is in the fit

Per epoch

- full affine transformation, 6 parameters: translation, rotation, scale, shear

Per source

- position (x_0, y_0) and proper motion (μx, μy) — 4 parameters

Detrending terms

- **differential chromatic refraction (DCR)** — `[1, sin(pa)·secz, cos(pa)·secz]` plus higher orders in `secz·sin(pa)` and `secz·cos(pa)`, fitted **in 6 colour

bins** (equal population). *This is included.*

- **annual term**
- **pixel-phase correction**, from the per-epoch registration shifts
- **SysRem**, 2 components on the astrometric residuals, computed **once over the

whole decade** and reused by every sub-fit

- iterative reweighting from the residuals, with moving-median outlier rejection

Convergence. 15 weighted iterations before SysRem, 6 after. The last three

iterations move the bright-star median by 0.003 mas (BLG41) and 0.001 (BLG01).

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3. What is not in the fit

- **parallax** — bulge sources give ≈0.12 mas, far below the noise floor
- **any microlensing or astrometric-deviation model** — the solution is model-free;

no theoretical curve of any kind has been fitted or subtracted

- **absolute astrometry** — there is no Gaia tie. Proper motions are *relative*,

in the registered pixel frame; 400 mas/pix is assumed, not measured, and the

orientation on the sky is unknown

- acceleration or binary motion
- any correction for the faint-end photometric bias (§6)

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4. Steps

1. **`KMT_pipelineI`** — registration, PSF photometry, source matching →

`MatchedSources`; cross-match to the OGLE catalogue for I and V, using a

per-field pixel offset (BLG41 229.202/229.283, BLG01 227.542/229.689).

2. **`ml.util.mmsFromMatchedSources`** — magnitudes rebuilt from `FLUX_PSF` (the

pipeline leaves every `MAG_*` column empty), per-epoch zero point anchored on

OGLE I, then SysRem photometry; airmass and parallactic angle computed from

JD and the field centre; colour from OGLE V−I; quality cuts.

3. **Joint fit 2016–2025** → proper motions. The event season is excluded so that it cannot help define the motion it is later measured against.

4. **Full-decade fit** with those motions held → one SysRem correction for the run.

5. **Ten single-season fits** reusing that correction.

6. **Final global fit, 2016–2026, motions solved.** *This is the solution behind every plot in this report.*

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5. Calibrating stars — selection, and their colour in the plots

Selection (all conditions required):

- **14 < I_OGLE < 19**
- **no companion brighter than I = 18 within 2.0 arcsec**, i.e. **5 KMT pixels**

at 0.4″/pix. The companion list is the OGLE catalogue, which is at 0.26″/pix and resolves about six times more sources than KMT detects, mapped into KMT pixels with the per-field offset above.

- a star's own OGLE counterpart is excluded from its companion test.

Surviving: **287 stars in BLG41, 253 in BLG01.**

Colour key in the RMS plots

colour	meaning
grey dots	all field stars
blue circles	**calibrating stars**, as selected above
black line	running median of *all* stars, in 0.5 mag bins
red star	the target

One point to be clear about: in this solution the per-epoch transformation is fitted from **all** sources, each weighted by its own residual scatter. The calibrating stars are a clean, well-measured subset shown for reference; they do not exclusively define the frame. (`UseRefSources`, which would restrict the frame fit to them, is available but is **off** here.)

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6. Caveats that matter for reading the plots

The measured magnitudes run faint at the faint end. `KMT − I_OGLE` is flat to ±0.04 down to I = 17, then rises to **+0.50 by I = 18.75**. The target's own measured magnitude is 18.70 against its catalogue value of **18.13**. All plots here therefore use the **OGLE catalogue I** on the magnitude axis. PSF and aperture photometry bracket OGLE and disagree by ~0.96 mag at I = 18.5, which points to crowding: OGLE lists 3 783 sources in this cut-out where KMT detects 594. **The astrometry is not affected** — positions come from the PSF centroid, not the flux, and removing the photometric anchor entirely was measured to cost 0.012 mas.

The CSV carries two error options; choose deliberately.

column	what it is	value
`errX_decade`, `errY_decade`	the black running-median curve of the RMS plot read at the target's catalogue magnitude I = 18.13, over the whole run. **Constant.**	21.80 / 23.48 (BLG41), 22.68 / 22.63 (BLG01)
`errX_season`, `errY_season`	the target's **own** residual RMS within its observing season. Follows its brightness.	11.4-33.8, see below

Neither is a per-epoch measurement uncertainty. The decade value is a population estimate at a fixed magnitude; the season value is the target's own scatter.

They differ by a factor of two where it matters most. The target is at I ≈ 18.1 for nine seasons and is magnified by about 1.1 mag in 2026, so:

season	target mag	errX / errY, BLG41	BLG01
1 (2016)	18.73	19.28 / 18.22	19.32 / 19.51
4 (2019)	18.71	31.92 / 26.31	29.21 / 26.56
6 (2022)	18.74	33.81 / 27.17	38.55 / 31.45

9 (2025)	18.61	16.85 / 15.13	16.76 / 14.75
10 (2026)	**17.65**	**12.46 / 11.38**	**11.82 / 11.54**
decade constant	—	21.80 / 23.48	22.68 / 22.63

Using the constant would overstate the uncertainty during the event by about a factor two, and understate it in 2022 by a third.

Plotted positions are relative to the target's own mean position, not to the field centre and not to any absolute reference. Each motion curve has the target's decade-average position subtracted, so the zero line is that average. The frame itself is set by the ensemble of 594 (BLG41) and 621 (BLG01) field stars through the per-epoch transformation, not by the geometric centre of the cut-out: differences along a curve are meaningful, the absolute level is not, and the level cannot be compared between the two fields, whose cut-outs have different origins and independently fitted frames. For scale, the target lies +0.42, +0.39 pix (+168, +155 mas) from the centre of the BLG41 cut-out and +1.55, +0.15 pix (+620, +60 mas) from the centre of BLG01; the centring absorbs both.

Only sources with an OGLE counterpart appear in the RMS plots (540 of 594 in BLG41, 478 of 621 in BLG01), which biases the plotted sample slightly bright.

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7. Where the numbers stand

	BLG41	BLG01
bright-star residual ($I < 17$), $\Delta X/\Delta Y$	6.414 / 6.879 mas	6.475 / 7.007 mas
calibrating stars, $\Delta X/\Delta Y$	8.29 / 8.30	7.47 / 8.14
target, whole decade	24.62 / 21.00	25.48 / 21.78
target, 2026 season only	≈ 11.7 / 11.1	≈ 12.2 / 11.7

The target's decade figure is dominated by the nine pre-event seasons where it sits at $I \approx 18.1$; in 2026 the event brightens it by about 1.1 mag and its residual halves. **12 mas is the target's 2026 precision, 24 mas its decade-average precision** — they are different quantities.

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8. Files in this report

file	contents
`report_RMS_afterPM.png`	residual RMS against OGLE I, per axis, per field — **after position and proper motion removed**
`report_motion___nobin.png`	source motion, unbinned; top = position with PM **retained**, bottom = residual
`report_motion___binsid.png`	the same, binned in **one sidereal month** (27.321661 d); error bars are the **RMS within the bin**
`source_motion_.csv`	`JD, X_mas, Y_mas, errX_decade, errY_decade, errX_season, errY_season, season`

Eight motion figures in total: 2 fields × 2 axes × 2 binnings.

Figures

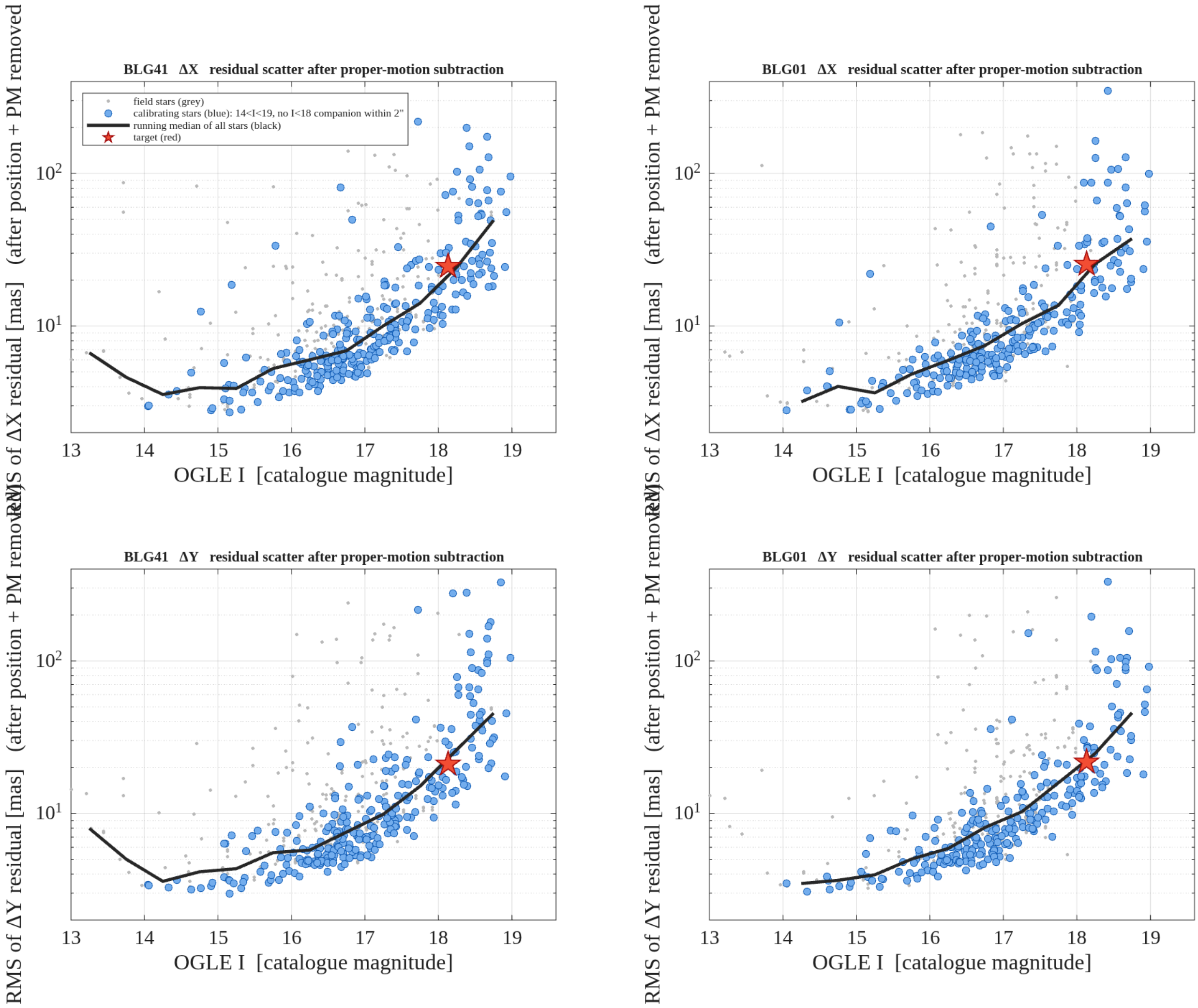

Figure 1. Residual RMS against the OGLE catalogue magnitude, per axis and per field, **after position and proper motion have been removed**. Grey: all field stars. Blue: calibrating stars (14<I<19, no I<18 companion within 2 arcsec). Black: running median of all stars, the curve the CSV decade errors are read from. Red: the target.

`report_RMS_afterPM.png`

Figure 2. **BLG41, X** — unbinned, one point per exposure, no error bars. Top: the measured X position with the proper motion **retained**. Bottom: the residual after position and proper motion are removed. Positions are relative to the target's own mean.
`report_motion_BLG41_X_nobin.png`

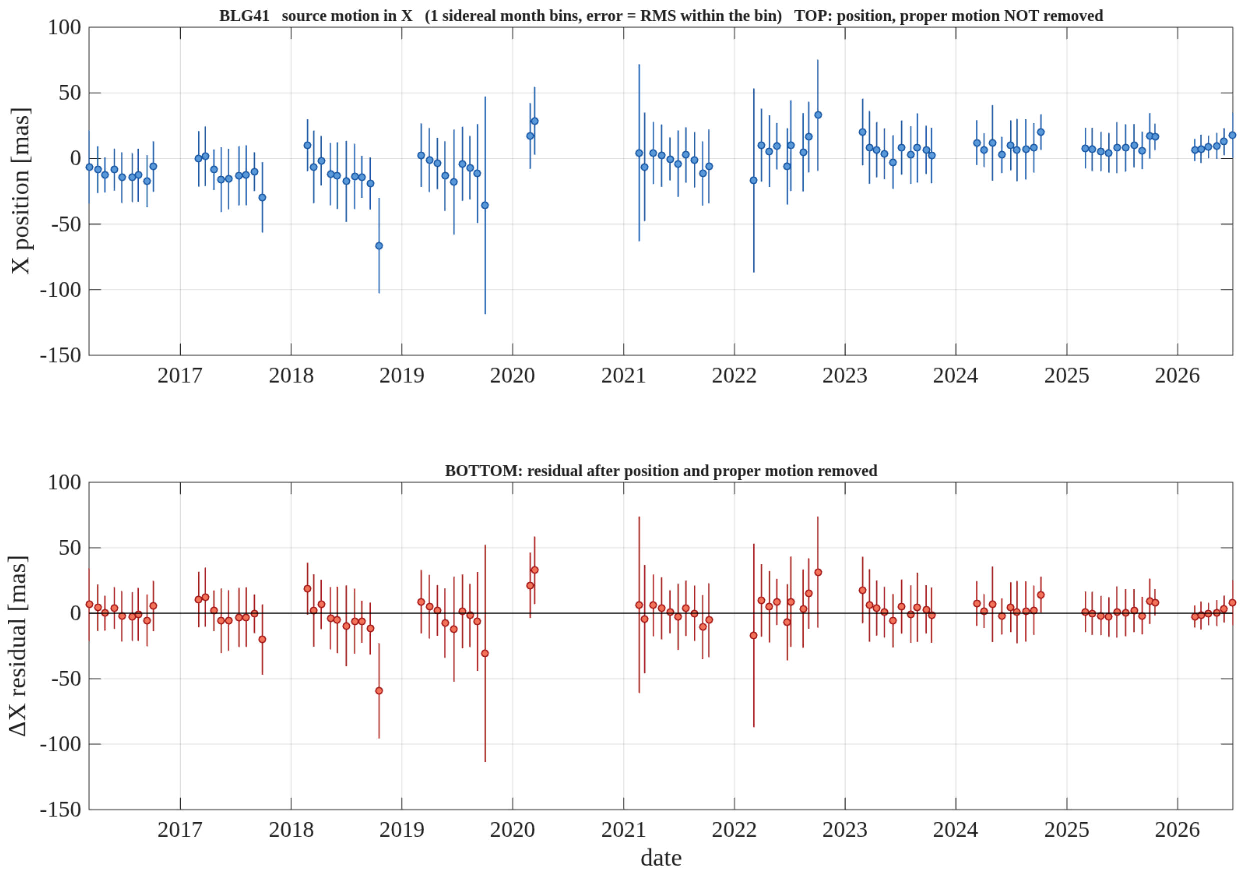

Figure 3. **BLG41, X** — binned in one sidereal month (27.321661 d); error bars are the RMS within the bin. Top: the measured X position with the proper motion **retained**. Bottom: the residual after position and proper motion are removed. Positions are relative to the target’s own mean.

report_motion_BLG41_X_binsid.png

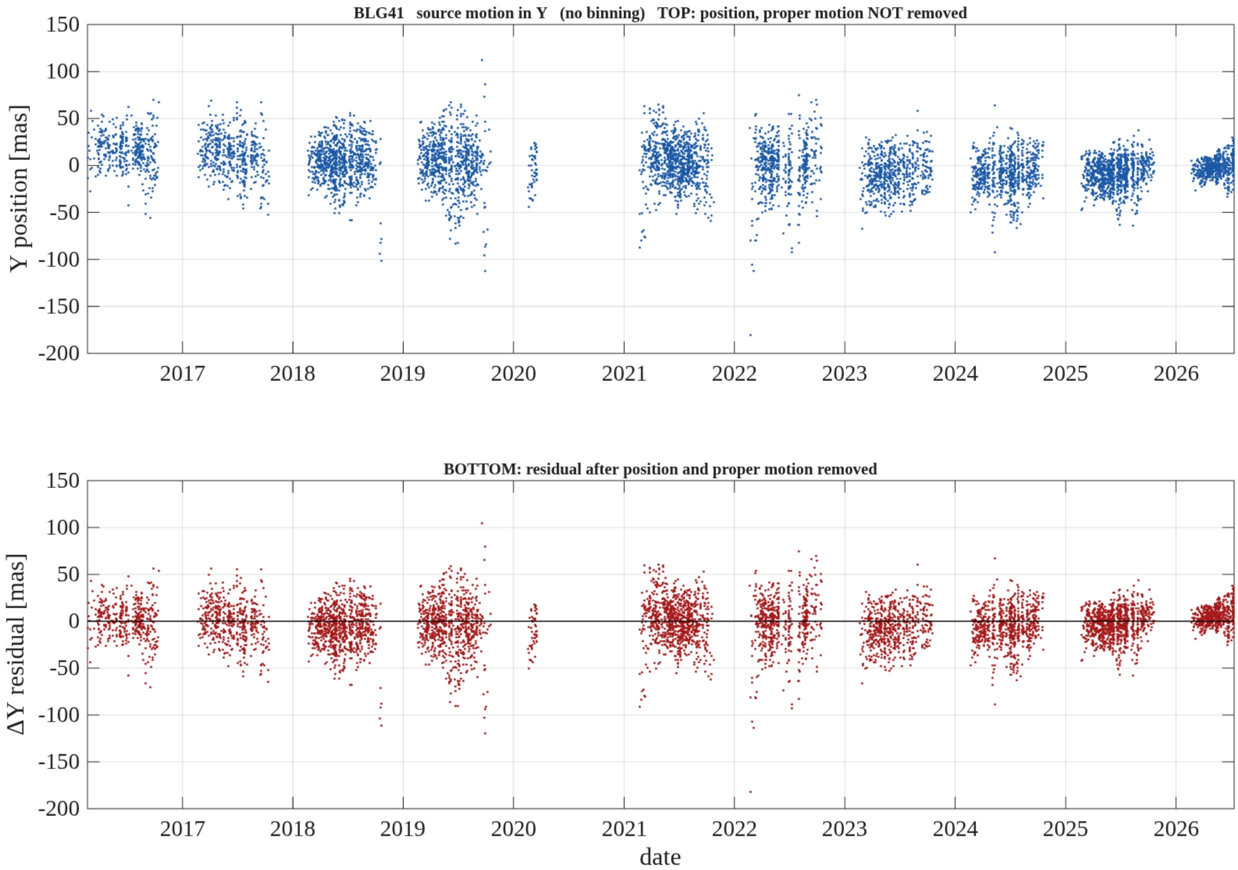

Figure 4. **BLG41, Y** — unbinned, one point per exposure, no error bars. Top: the measured Y position with the proper motion **retained**. Bottom: the residual after position and proper motion are removed. Positions are relative to the target’s own mean.

report_motion_BLG41_Y_nobin.png

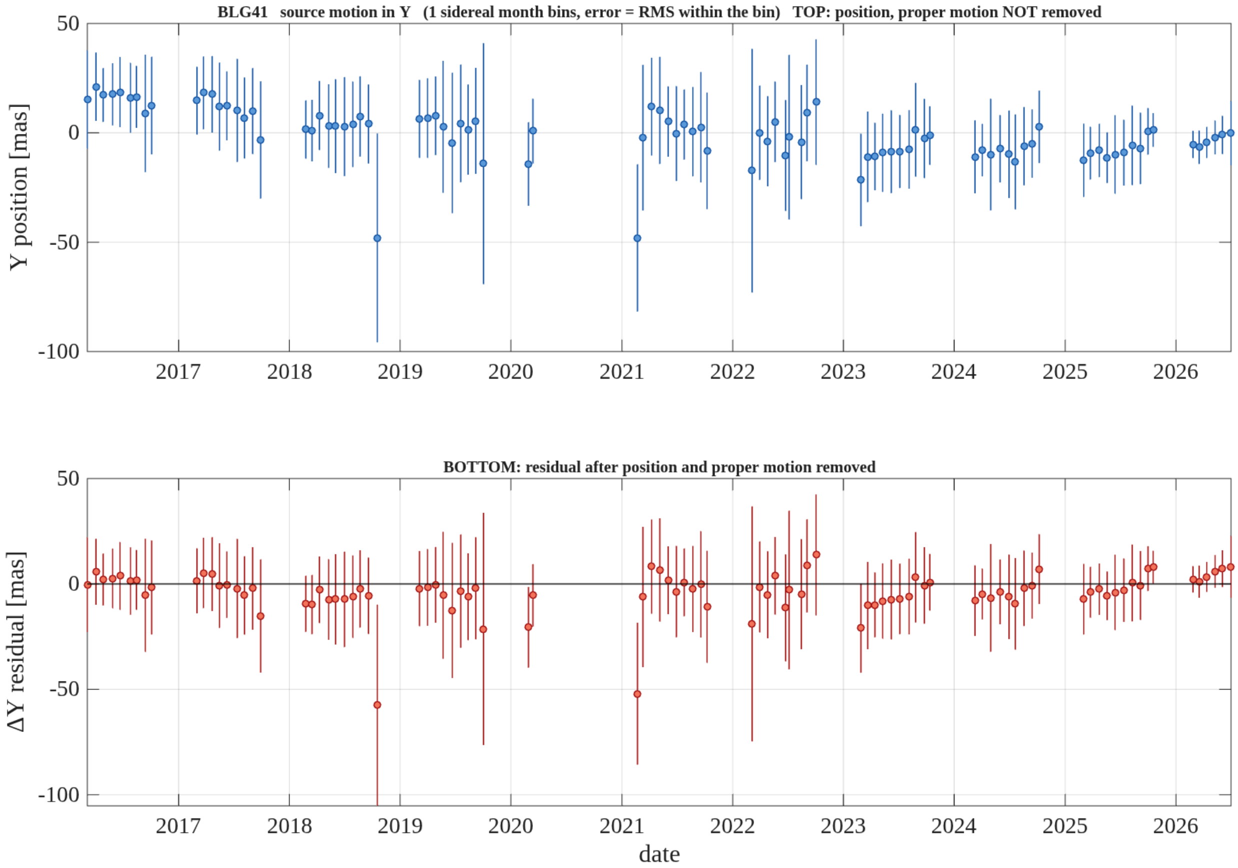

Figure 5. **BLG41, Y** — binned in one sidereal month (27.321661 d); error bars are the RMS within the bin. Top: the measured Y position with the proper motion **retained**. Bottom: the residual after position and proper motion are removed. Positions are relative to the target’s own mean.

`report_motion_BLG41_Y_binsid.png`

Figure 6. **BLG01, X** — unbinned, one point per exposure, no error bars. Top: the measured X position with the proper motion **retained**. Bottom: the residual after position and proper motion are removed. Positions are relative to the target's own mean.
`report_motion_BLG01_X_nobin.png`

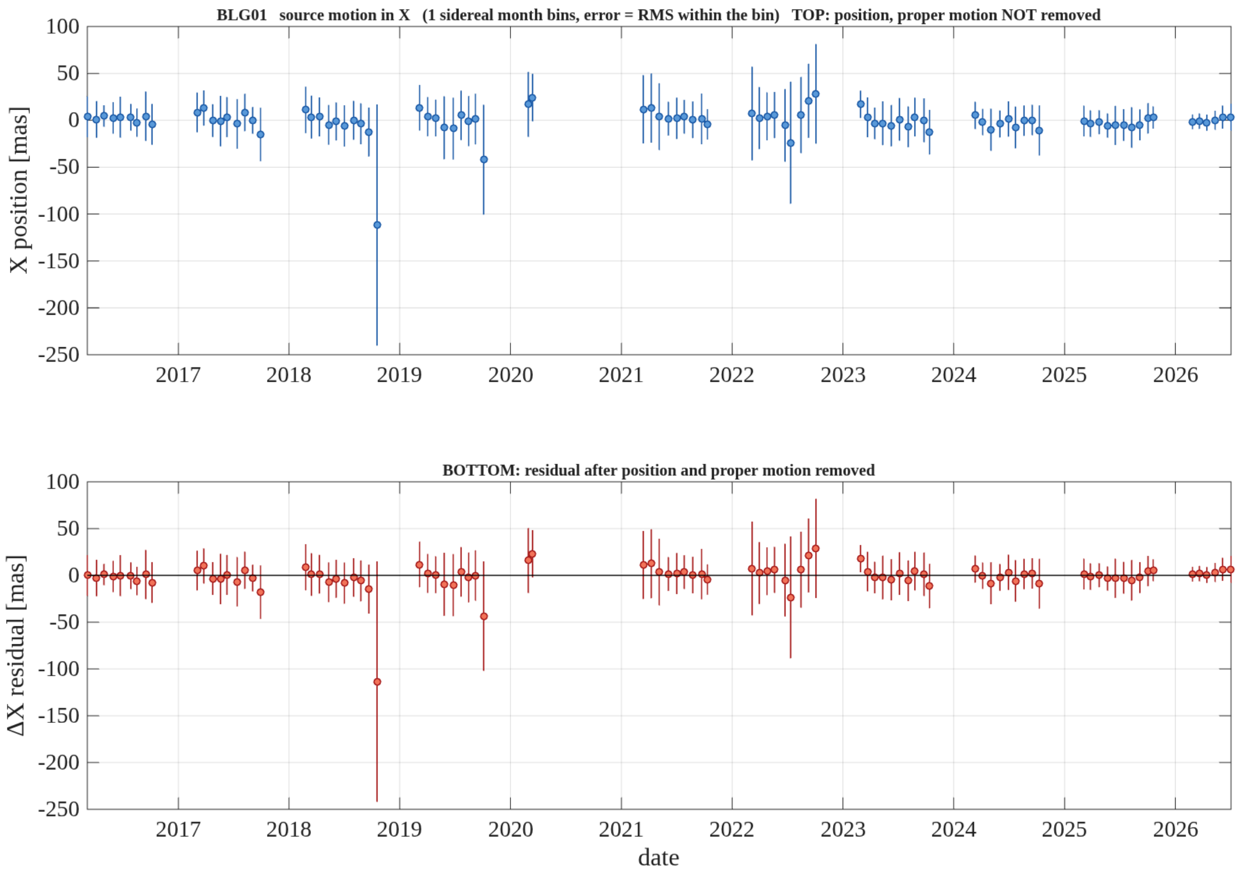

Figure 7. **BLG01, X** — binned in one sidereal month (27.321661 d); error bars are the RMS within the bin. Top: the measured X position with the proper motion **retained**. Bottom: the residual after position and proper motion are removed. Positions are relative to the target's own mean.
`report_motion_BLG01_X_binsid.png`

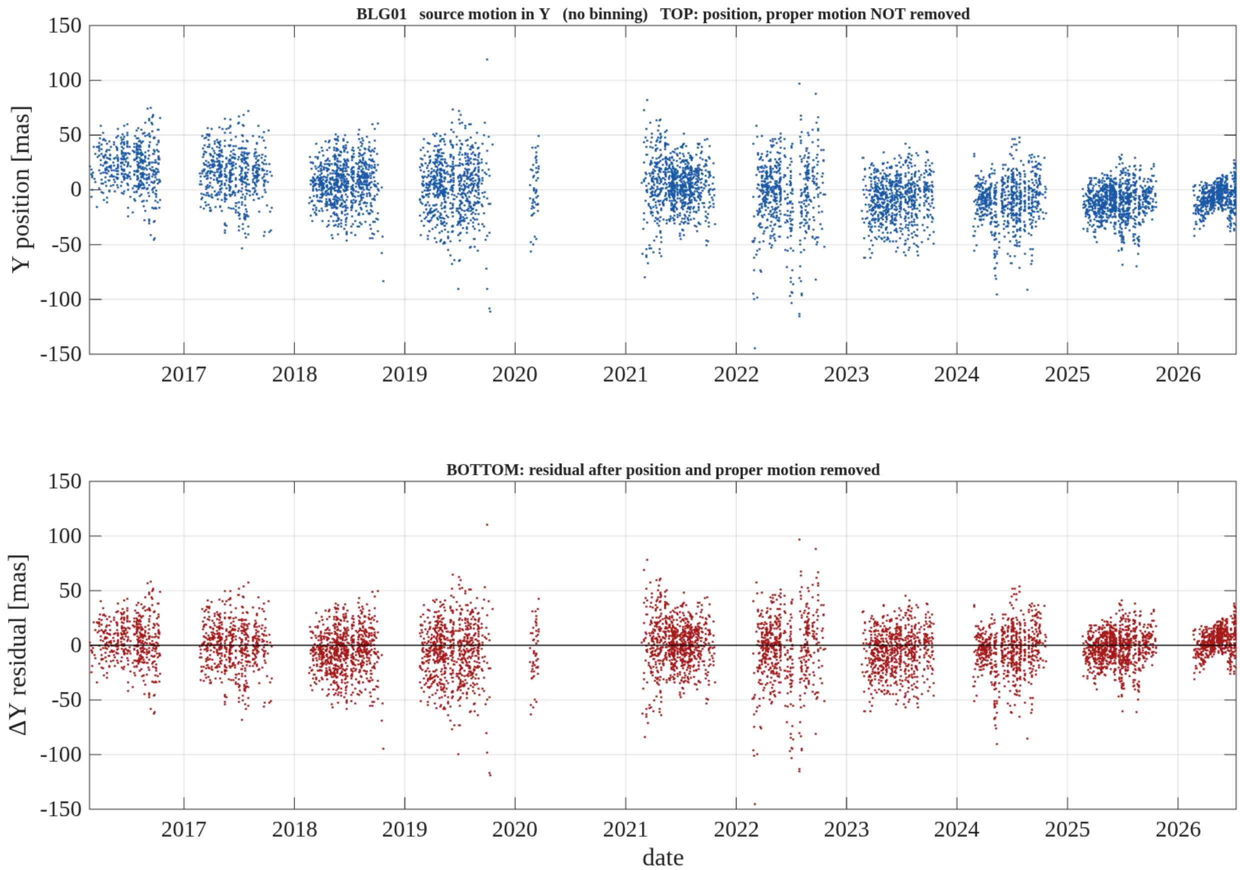

Figure 8. **BLG01, Y** — unbinned, one point per exposure, no error bars. Top: the measured Y position with the proper motion **retained**. Bottom: the residual after position and proper motion are removed. Positions are relative to the target's own mean.
`report_motion_BLG01_Y_nobin.png`

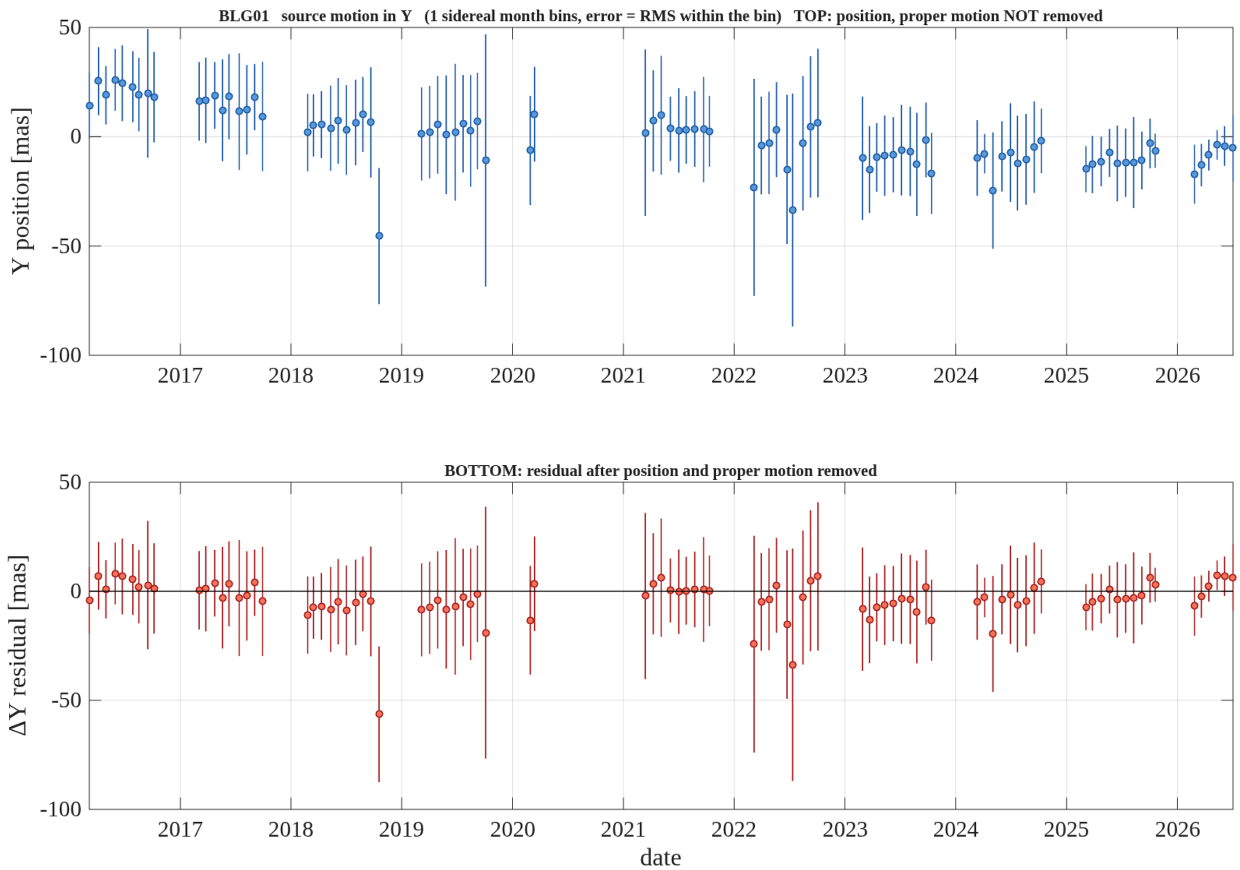

Figure 9. **BLG01, Y** — binned in one sidereal month (27.321661 d); error bars are the RMS within the bin. Top: the measured Y position with the proper motion **retained**. Bottom: the residual after position and proper motion are removed. Positions are relative to the target's own mean.

`report_motion_BLG01_Y_binsid.png`